View Meta-Reviews

Paper ID

561

Paper Title

On the Complexity of Offline Reinforcement Learning with Q*-Approximation and Partial Coverage

META-REVIEWER #1

META-REVIEW QUESTIONS

2. Meta-review

While the reviewers agree that there exist some interesting results contained within, all reviewers also agree that at the current level, the paper is very poorly organized, to such an extent that it makes it difficult to read and get a good understanding of what is going on. While I would not normally allow this to unduly influence my decision, after reading over the paper, I agree with the reviewers. I also think that the technical issue with Lemma 10 pointed out by one reviewer, while potentially fixable in the manner described by the authors, remains an issue, as the poor organization of the paper as it stands makes it difficult to discern precisely what the authors wish to prove or why they wish to prove it. I agree with two of the reviewers regarding the potential interest in the lower bound, and I encourage the authors to submit this work after substantial reorganization and editing.

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Paper Title

On the Complexity of Offline Reinforcement Learning with Q*-Approximation and Partial Coverage

Reviewer #1

Questions

1. Rating (scale of 1-7)

3: Marginally below acceptance threshold

2. Confidence (scale of 1-4)

2:It is possible that the reviewer did not understand important parts of the paper

3. Review

The paper studies the question of whether offline reinforcement learning is statistically tractable under Bellman completeness and partial coverage assumptions. This is a somewhat niche but important technical problem in RL theory; prior work has shown that pessimism is a useful algorithmic primitive for handling partial coverage, but implementing pessimism successfully typically relies on stronger function approximation assumptions than just bellman completeness. This paper starts with an information theoretic lower bound, showing that one cannot achieve sample efficiency in this setting. From here the paper turns to upper bounds, using a framework inspired by the DEC, to obtain new offline RL algorithms for a variety of settings.

Personally I found the paper pretty hard to read. The main lower bound is pretty cool and I get the intuition, but the upper bound sections (4-onward) are quite disorganized and I lose the connective thread between the various subsections. I am not sure what the concrete end-to-end result is from these section. Honestly I think it is quite likely there is something interesting going on, but I am not able to see what the main takeaway is, how that compares to prior work, and why it is in interesting.

References:

Zanette, Wainwright, Brunskill. Provable benefits of actor critic methods for offline

reinforcement learning. 2021 -- This paper seems pretty relevant although they are making stronger assumptions.

Reviewer #2

Questions

1. Rating (scale of 1-7)
3: Marginally below acceptance threshold

2. Confidence (scale of 1-4)
2:It is possible that the reviewer did not understand important parts of the paper

3. Review
Summary
This paper studies offline reinforcement learning under partial coverage with Q*-realizability and Bellman completeness. There are a few main contributions: an information-theoretic lower bound; a policy-centric decision criterion and offline decision rule (E2D.OR); gap- and curvature-dependent guarantees in both unregularized and regularized settings; extensions to low Q-Bellman-rank settings. The problem is important, and the distinction between policy-centric and value-centric pessimism is a meaningful conceptual contribution.

I have two techinical concerns:
- In lemma 10, we proved the signed weight BE is bounded, how does this imply the boundness of the squared BE?
- In trajectory sample lower bound, I think one should show that the full trajectory observation model is no more informative than the earlier tuple-based hard instance.

Overall, I think the paper is trying to be ambitious and cover a few results in different directions under the loosely-connected theme. While each result individually is interesting to a certain degree, I am not fully convinced that compiling them together simply makes the paper stand out. Some results, e.g., the analysis of CQL, does not really add anything to the paper or even intersting on its own.

Reviewer #3

Questions

1. Rating (scale of 1-7)
4: Marginally above acceptance threshold

2. Confidence (scale of 1-4)
3: The reviewer is confident in their evaluation

3. Review
Post rebuttal:

After reading the other reviews and the rebuttal, I think some of my concerns around the gap issue have been addressed. I agree with Reviewer #1's point that presenting the main result in a more clear way will improve the paper. I had missed the technical concern that Reviewer #2 points out about Lemma 10 in my initial review. While the authors claim they have addressed this in the rebuttal it would have been nice to see a complete version before accepting the paper. I still think there are good contributions and most of the presentation surrounding the technical details is good. However, I do appreciate that the paper will benefit from a more clear presentation of the main results and fixing the technical issue with Lemma 10.

This work studies the Offline RL problem with function approximation under partial coverage assumptions.

The main contributions of the work are as follows. First the authors show that in addition to Q*-realizability and Bellman completeness the sample complexity bounds necessarily incur a value-gap type quantity. This is somewhat surprising as it requires a sufficiently easy to distinguish unique optimal action at every state. Something that is unclear to me is if the min in the definition of the value gap needs to be taken over all states or just states in the support of \pi*.

Next, the authors propose an algorithm (Algorithm 2) based on a decision-estimation coefficient. The authors analyze two versions of this algorithm, one with a ratio version of the DEC and one with an offset version. These DECs are in turn bounded by explorability ratios which are then used to finally bound the sample complexity in terms of a gap-dependent quantity, which is the easiest to interpret. The tuning of the offset version of the algorithm is agnostic to the gap-dependent quantity, however, it has linear dependence on partial coverage coefficient. The sub-optimality bound for the ratio version of the algorithm has sub-linear dependence on the coverage coefficient, however, it also depends on a quantity that's hard to control or reason about in terms of the sub-optimality gap of \hat f. The authors also present a different type of pessimistic algorithm (Algorithm 1) and derive a similar sub-optimality bound.

In Section 6 the authors show how to control the estimation error due to the confidence sets used in Algorithm 1 and Algorithm 2 and how to construct good confidence sets.

Lemma 11 seems to be the strongest result requiring somewhat weak assumptions such as small Bellman rank and policy feature coverage. The double policy sampling assumption also seems reasonable to me.

In Section 7 the authors briefly discuss CQL and show a sub-optimality guarantee under Q*-realizability, Bellman completeness and admissibility of the sampling distribution.

The presentation of the work is good, the authors give ample intuition behind the lower bound construction and how to interpret the sub-optimality guarantees. The notation is a bit heavy but I don't see a good way to improve this too much.

One potential weakness of this work is related to my comment/question on the definition of the value-gap in the lower bound. While all bounds in this work use the partial coverage coefficient, the value-gap quantities that appear in most places involve a quantity computed over all states, rather than just states in the support of \pi*. Is it possible to tighten the value-gap definition to only depend on states in the support of \pi*, or even better, to depend on some average notion or expected value gap, similarly to what has been considered in prior work in the online setting for optimistic regret bounds, e.g., [1] and [2].

Another potential weakness is the computational complexity of the proposed algorithms. CQL seems to be the most computationally attractive algorithm as it does not need to solve a min-max-min problem, however, it is unclear how the algorithm depends on a value-gap type quantity.

In general, I like the contributions of this work and there seems to be some good novelty in the lower bound constructions. I am not sure how novel the upper bound techniques are, however.

[1] Wagenmaker, Andrew J., Max Simchowitz, and Kevin Jamieson. "Beyond no regret: Instance-dependent pac reinforcement learning." Conference on Learning Theory. PMLR, 2022.
[2] Dann, Christoph, et al. "Beyond value-function gaps: Improved instance-dependent regret bounds for episodic reinforcement learning." Advances in Neural Information Processing Systems 34 (2021): 1-12.